Shaden **Shaar**

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About Me

A final-year PhD candidate in Computer Science at Cornell University, advised by Prof. Claire Cardie. My thesis, *Long-Form Input Understanding across Vision-and-Language Tasks*, asks what a model needs from a long input: **what it has to locate, and when locating the right piece is not enough**. I have studied this question in text (TACL 2025) and in short video, for which I built the *MovieRecapsQA* benchmark (CVPR 2026). I am now investigating **long narrative videos**. In parallel, I collaborate with NewYork-Presbyterian Hospital on **heart-failure** projects, ranging from analyzing UNOS exception requests (JHLT 2026) to predictive medication models for heart-failure patients. Previously, at the Qatar Computing Research Institute, I worked with Prof. Preslav Nakov and Prof. Giovanni Da San Martino on **news fact-checking and propaganda detection**.

Research Interests

- **Long-form input understanding.** What models need from long inputs across text documents, short video, and full-length film: the kind of localization a task requires, and the kind of knowledge it draws on.
- **Benchmarks and evaluation for open-ended generation.** Building the datasets that make model failures observable, and evaluating answers no fixed reference can capture.
- **Clinical NLP.** Using LLMs to surface clinical decision patterns from unstructured medical narratives, with a focus on heart failure and heart transplant care.
- **News fact-checking and propaganda detection** *(prior).* Detecting previously fact-checked claims, and identifying propaganda and persuasion techniques across text, images, and memes.

Highlights

- **h-index 24** with ∼2,858 citations; 32 publications across **CVPR**, **TACL**, **ACL**, **EMNLP**, **NAACL**, **COLING**, **IJCAI**, **ICWSM**, and **LREC**.
- Visiting the **Berkeley NLP group**, UC Berkeley (Fall 2026), working with Dr. Sewon Min.
- Gave a talk at the **Berkeley NLP retreat** (2026).
- Created ***MovieRecapsQA*** (CVPR 2026), an open-ended benchmark for question answering over full-length film.
- **University Fellowship**, Cornell University (2021), awarded for exceptional preparation and promise.
- Co-organized shared tasks at **CLEF–CheckThat!** (2020, 2021), **SemEval-2021** Task 6 on persuasion in memes, and **NLP4IF-2021**.
- Introduced the task of **detecting previously fact-checked claims** (ACL 2020, 240+ citations), now a standing task in the CLEF–CheckThat! lab and adopted by fact-checking and news organizations.
- **Best Demo Award (Honorable Mention)**, ACL 2020, for *Prta*, a system for analyzing propaganda techniques in the news.

Education

Cornell University — *Ithaca, NY, USA*

PhD in Computer Science — *Aug. 2021 – Present*

Minor: Applied Mathematics.

Cornell University — *Ithaca, NY, USA*

M.S. in Computer Science — *Aug. 2021 – May 2024*

Carnegie Mellon University — *Doha, Qatar*

B.S. in Computer Science — *Aug. 2015 – May 2019*

Minor in Mathematics; College and University Honors

Work Experience

Zillow Group — *Remote, USA*

Applied Scientist Intern — *May 2025 – Aug. 2025*

- Built a home-purchase co-pilot: a multi-turn assistant that answers buyer questions, tracks purchase intent as it accumulates over a conversation, and narrows a large listing inventory toward properties that fit.
- Post-trained the dialogue policy on long multi-turn conversation data, running DPO over preference pairs scored on whole dialogues, then GRPO with a trajectory-level reward so credit lands on the conversation rather than the individual turn.
- Built the reward-modeling and evaluation pipeline behind that training, including rollout generation and conversation-level scoring for grounded property-search dialogue.

ScaleAI, Inc. *New York City, NY, USA*

Machine Learning Research Engineer Intern *Jan. 2025 – May 2025*

- Built the judge co-pilot for the Qatari judicial department: a set of case-analysis services, each exposed as an API, deployed together as a judge-facing dashboard.
- Designed the Arabic information-extraction pipeline, running OCR over scanned filings, then event extraction that links each extracted event to the evidence spans supporting it in the case document, feeding a case summarizer built on that structure.
- Built precedent lookup as a RAG system over constitutional bylaws and prior Qatari Supreme Court cassation rulings, combining dense embedding retrieval with sparse lexical matching and reranking the merged candidate set.

Apple *Seattle, WA, USA*

AI/ML Research Intern *May 2022 – Aug. 2022*

- Built a conditional synthetic dialogue generator for Siri training with Dr. Alex Churchill, producing multi-turn conversation data for scenarios with little or no real usage data to draw on.
- Simulated user behaviour to generate the human side of each conversation, so the pipeline produced complete multi-turn dialogues rather than isolated utterances.
- Conditioned generation on intent and user behaviour, letting a target dataset be specified to arbitrary requirements rather than freely sampled, and ran zero-shot without seed dialogues for the target scenarios.

Qatar Computing Research Institute *Doha, Qatar*

Research Assistant *Jul. 2019 – Aug. 2021*

- **Automated fact-checking:** introduced the task of detecting previously fact-checked claims, matching an incoming claim against a database of prior verdicts so a checker could reuse existing work instead of starting over. Follow-up work extended it to document-level matching and to the role of surrounding context, with papers at ACL, EMNLP, NAACL, and IJCAI.
- **Propaganda and persuasion detection:** built *Prta*, a public system that flags propaganda techniques in news articles (ACL 2020 Best Demo, Honorable Mention), then extended the work to multimodal memes where the technique is split across text and image, published at ACL and SemEval.
- **COVID-19 misinformation:** built pipelines for factuality, harmfulness, and framing of pandemic claims across multiple languages and social platforms, with papers at ICWSM, EMNLP, RANLP, and COLING.
- **Emotion detection:** cross-lingual classification for low-resource languages with little annotated data (LREC 2022).
- Co-organized shared tasks at SemEval, CLEF–CheckThat!, and ACL workshops, releasing the datasets and evaluation setups these problems are still benchmarked on. Advised by Dr. Preslav Nakov and Prof. Giovanni Da San Martino.

Publications

Journals

1. Frye, J., **Shaar, S.**, Cardie, C., DeFilippis, E., Estrin, D., Sayer, G., Uriel, N., et al. "Thematic Analysis of Accepted Exception Requests for Heart Transplant Candidates Using a Large Language Model." *The Journal of Heart and Lung Transplantation* 45(5):460, 2026. **(JHLT'2026)**.
2. **Shaar, S.**, Chen, W., Chatterjee, M., Wang, B., Zhao, W., and Cardie, C. "Are *Triggers* Needed for Document-Level Event Extraction?" *Transactions of the Association for Computational Linguistics* **(TACL'2025)**.

Conferences

1. **Shaar, S.**, Thymes, B., Chaixanien, S., Cardie, C., and Hariharan, B. "MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark." *Conference on Computer Vision and Pattern Recognition* **(CVPR'2026)**.
2. Chaixanien, S., Choi, E., **Shaar, S.**, and Cardie, C. "Pungene at DialAM-2024: Identification of Propositional and Illocutionary Relations." *Workshop on Argument Mining at ACL 2024* **(ArgumentMining@ACL'2024)**.
3. Aydin, A. N., **Shaar, S.**, and Cardie, C. "Edward Said at Touché: Human Value Detection Using Transformers and Upsampling." *Working Notes of the Conference and Labs of the Evaluation Forum* **(CLEF'2024)**.
4. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "The CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *European Conference on Information Retrieval* **(ECIR'2022)**.
5. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "Overview of the CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *Cross-Language Evaluation Forum for European Languages* **(CLEF'2022)**.
6. **Shaar, S.**, Georgiev, N., Alam, F., Da San Martino, G., Mohamed, A., and Nakov, P. "Assisting the Human Fact-Checkers: Detecting All Previously Fact-Checked Claims in a Document." *Findings of the Association for Computational Linguistics: EMNLP 2022* **(EMNLP'2022)**.
7. **Shaar, S.**, Alam, F., Da San Martino, G., and Nakov, P. "The Role of Context in Detecting Previously Fact-Checked Claims." *Findings of the Association for Computational Linguistics: NAACL 2022* **(NAACL'2022)**.

8. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "A Second Pandemic? Analysis of Fake News About COVID-19 Vaccines in Qatar." *Findings of the Association for Computational Linguistics: NAACL 2022* **(NAACL'2022)**.
9. Hassan, S., **Shaar, S.**, and Darwish, K. "Cross-Lingual Emotion Detection." *Proceedings of the Thirteenth Language Resources and Evaluation Conference* **(LREC'2022)**.
10. Alam, F., Cresci, S., Chakraborty, T., Silvestri, F., Dimitrov, D., Da San Martino, G., **Shaar, S.**, Firooz, H., and Nakov, P. "A Survey on Multimodal Disinformation Detection." *International Conference on Computational Linguistics* **(COLING'2022)**.
11. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "Detecting Propaganda Techniques in Memes." *Association for Computational Linguistics* **(ACL–IJCNLP'2021)**.
12. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "SemEval-2021 Task 6: Detection of Persuasion Techniques in Texts and Images." *ACL-IJCNLP SemEval Workshop* **(SemEval@ACL-IJCNLP'2021)**.
13. **Shaar, S.**, Alam, F., Da San Martino, G., Nikolov, A., Zaghouani, W., Nakov, P., and Feldman, A. "Findings of the NLP4IF-2021 Shared Tasks on Fighting the COVID-19 Infodemic and Censorship Detection." *ACL NLP4IF Workshop* **(NLP4IF@ACL'2021)**.
14. Nakov, P., Da San Martino, G., Elsayed, T., Barrón-Cedeño, A., Míguez, R., **Shaar, S.**, Alam, F., Haouari, F., Hasanain, M., Babulkov, N., Nikolov, A., Shahi, G., Struß, J., and Mandl, T. "The CLEF-2021 CheckThat! Lab on Detecting Check-Worthy Claims, Previously Fact-Checked Claims, and Fake News." *European Conference on Information Retrieval* **(ECIR'2021)**.
15. Skuczynska, B., **Shaar, S.**, Spenader, J., and Nakov, P. "BeaSku at CheckThat! 2021: Fine-Tuning Sentence BERT with Triplet Loss and Limited Data." *CLEF Conference and Labs of the Evaluation Forum* 639–647, **(CLEF'2021)**.
16. Alam, F., **Shaar, S.**, Dalvi, F., Sajjad, H., Nikolov, A., Mubarak, H., Da San Martino, G., Abdelali, A., Durrani, N., Darwish, K., Al-Homaid, A., Zaghouani, W., Caselli, T., Danoe, G., Stolk, F., Bruntink, F., and Nakov, P. "Fighting the COVID-19 Infodemic: Modeling the Perspective of Journalists, Fact-Checkers, Social Media Platforms, Policy Makers, and the Society." *Findings of EMNLP* **(EMNLP'2021)**.
17. Alam, F., Dalvi, F., **Shaar, S.**, Durrani, N., Mubarak, H., Nikolov, A., Da San Martino, G., Abdelali, A., Sajjad, H., Darwish, K., and Nakov, P. "Fighting the COVID-19 Infodemic in Social Media: A Holistic Perspective and a Call to Arms." *International AAAI Conference on Web and Social Media* **(ICWSM'2021)**.
18. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "COVID-19 in Bulgarian Social Media: Factuality, Harmfulness, Propaganda, and Framing." *International Conference on Recent Advances in Natural Language Processing* **(RANLP'2021)**.
19. Nakov, P., Corney, D., Hasanain, M., Alam, F., Elsayed, T., Barrón-Cedeño, A., Papotti, P., **Shaar, S.**, and Da San Martino, G. "Automated Fact-Checking for Assisting Human Fact-Checkers." *International Joint Conferences on Artificial Intelligence* **(IJCAI'2021)**.
20. **Shaar, S.**, Da San Martino, G., Babulkov, N., and Nakov, P. "That Is a Known Lie: Detecting Previously Fact-Checked Claims." *Association for Computational Linguistics* **(ACL'2020)**.
21. Da San Martino, G., **Shaar, S.**, Zhang, Y., Yu, S., Barrón-Cedeño, A., and Nakov, P. "Prta: A System to Support the Analysis of Propaganda Techniques in the News." *Association for Computational Linguistics* **(ACL'2020)**. **Best Demo Award (Honorable Mention).**
22. **Shaar, S.**, Nikolov, A., Babulkov, N., Alam, F., Barrón-Cedeño, A., Elsayed, T., Hasanain, M., Suwaileh, R., Haouari, F., Da San Martino, G., and Nakov, P. "Overview of CheckThat! 2020 English: Automatic Identification and Verification of Claims in Social Media." *CLEF Working Notes* 2696, **(CLEF'2020)**.
23. Barrón-Cedeño, A., Elsayed, T., Nakov, P., Da San Martino, G., Hasanain, M., Suwaileh, R., Haouari, F., Babulkov, N., Hamdan, B., Nikolov, A., **Shaar, S.**, and Ali, Z. S. "Overview of CheckThat 2020: Automatic Identification and Verification of Claims in Social Media." **(CLEF'2020)**, 11:215–236.
24. **Shaar, S.**, Razak, S., Dalvi, F., and Moosavi, S. "Group Identification in Crowded Environments Using Proximity Sensing." *IEEE Conference on Local Computer Networks* **(LCN'2018)**.
25. Hassan, S., **Shaar, S.**, Raj, B., and Razak, S. "Interactive Evaluation of Classifiers Under Limited Resources." *IEEE International Conference on Machine Learning and Applications* **(ICMLA'2018)**.

Honors & Awards

2021	**University Fellowship,** Cornell University, awarded for exceptional preparation and promise	*Ithaca, NY, USA*
2020	**Best Demo Award (Honorable Mention),** ACL 2020, for *Prta: A System to Support the Analysis of Propaganda Techniques*	*Online*
2019	**College and University Honors,** Carnegie Mellon University, for outstanding cumulative GPA	*Doha, Qatar*
2019	**Dean's List (5 awards),** Carnegie Mellon University, F15, S16, F16, F17	*Doha, Qatar*
2015	**50% Academic Merit Scholarship,** Carnegie Mellon University, awarded to a select group of incoming students	*Doha, Qatar*

Academic Services

2021	**Co-organizer,** SemEval–2021 Task 6: Detection of Persuasion Techniques in Texts and Images	*Online*
2021	**Co-organizer,** CLEF–2021 CheckThat! Lab on detecting check-worthy and previously fact-checked claims	*Bucharest, Romania*
2020	**Co-organizer,** CLEF–2020 CheckThat! Lab on automatic identification and verification of claims	*Online*
2020	**Reviewer,** ACL, EMNLP, NAACL, EACL, COLING, AAAI, CIKM (Program Committee)	

Teaching Experience

Cornell University

Ithaca, NY, USA

Teaching Assistant

2021 – 2022

- CS 4740 – Natural Language Processing (Fall 2022, Lecturer: Prof. Lillian Lee).
- CS 5780 – Introduction to Machine Learning (Spring 2022).

Carnegie Mellon University

Doha, Qatar

Teaching Assistant

2016 – 2019

- 11-785 – Introduction to Deep Learning (Fall 2018, Spring 2019, Lecturer: Prof. Bhiksha Raj).
- 15-251 – Great Theoretical Ideas in Computer Science (Spring 2018, Spring 2019).
- 15-213 – Introduction to Computer Systems (Fall 2017).
- 15-112 – Fundamentals of Programming (Fall 2016, Fall 2017).
- 21-127 – Concepts of Mathematics (Spring 2016).